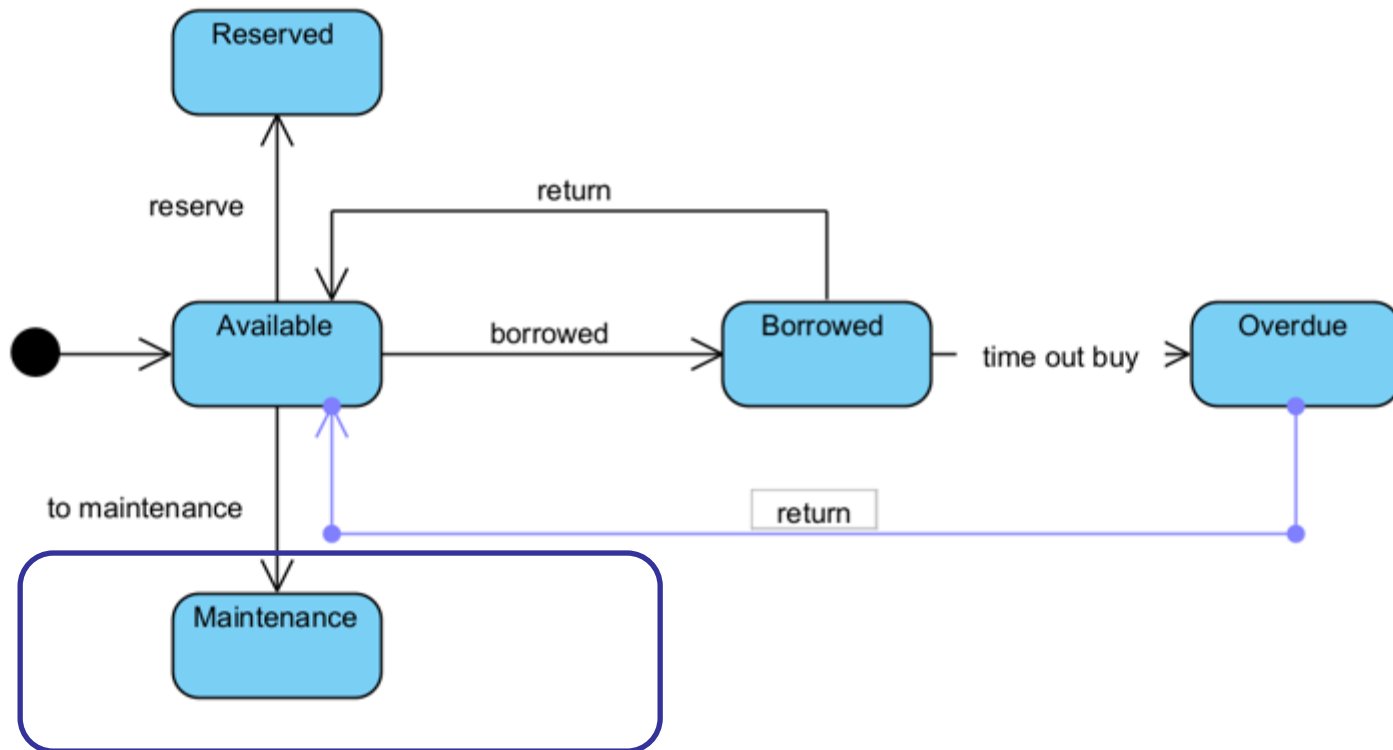


# Several Problems in the State Machine Modeling Experiment

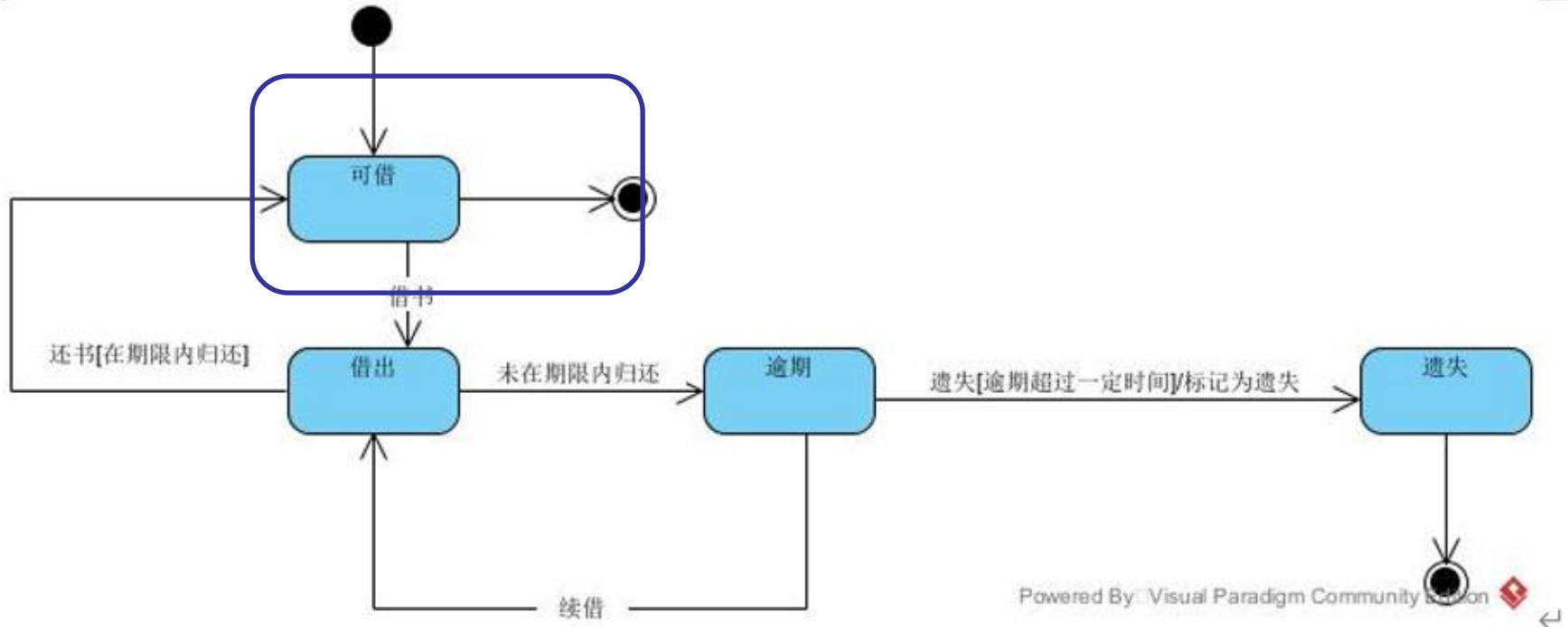
# What's inappropriate in the diagram??



# Suggestion

- Questioning the “Black-Hole” States: A black-hole state is one that has transitions into it but none out of it.

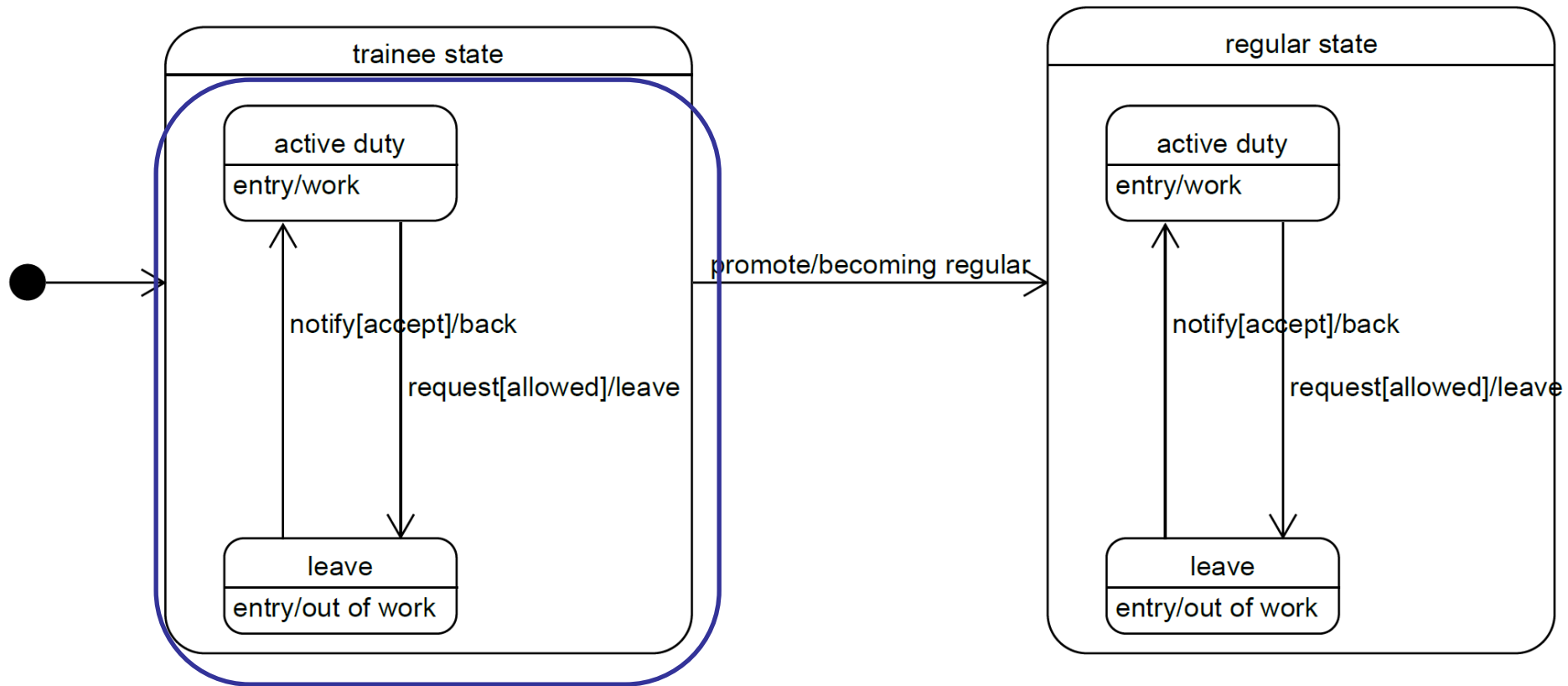
# What's inappropriate in the diagram??



# Suggestion

- Generally speaking, transitions between states should indicate the triggering events.

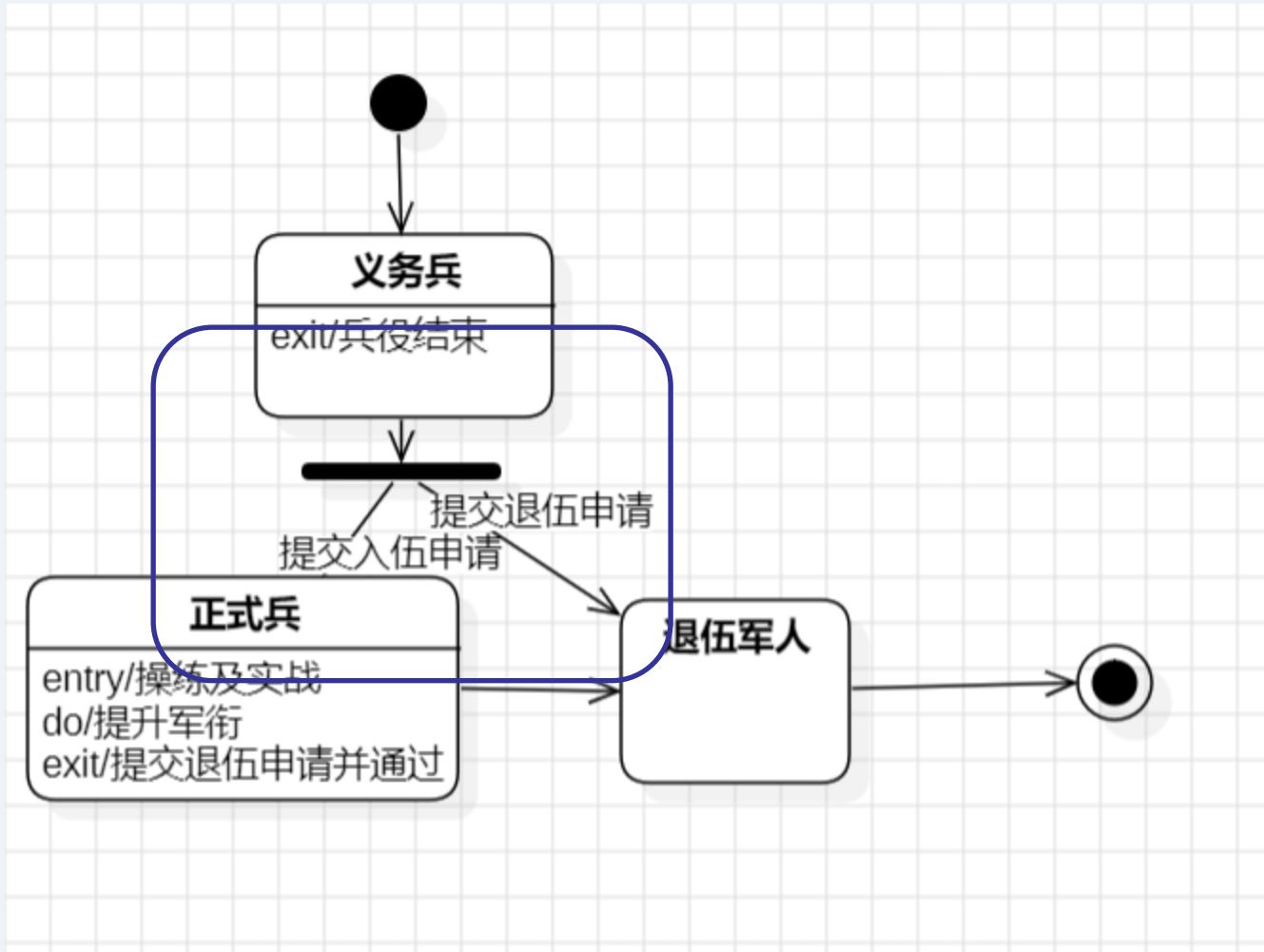
# What's inappropriate in the diagram??



# Suggestion

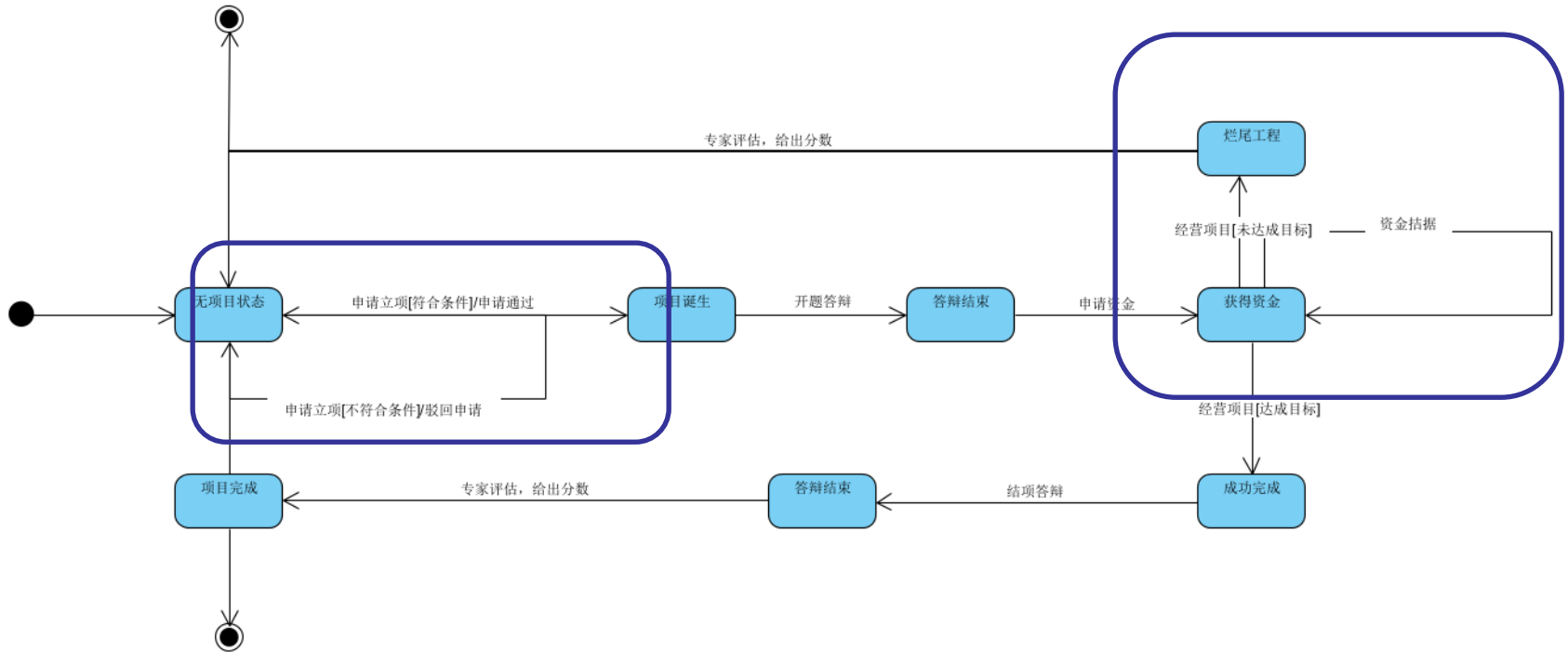
- In the sub-state machines within a state, it is also advisable to specify the initial state.

# What's inappropriate in the diagram??





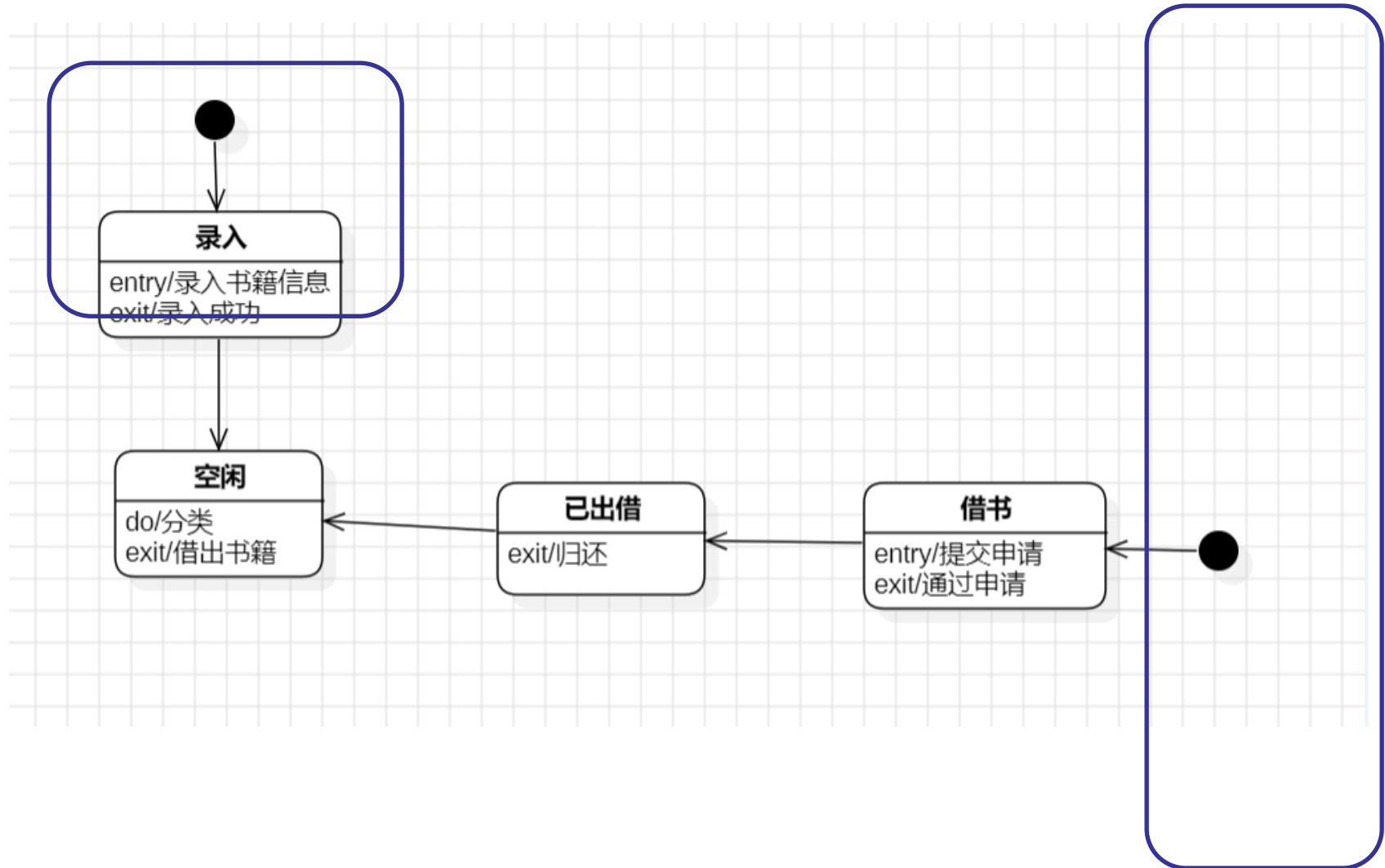
# What's inappropriate in the diagram??



# Suggestion

- In a state machine diagram, sometimes we can use the Choice Pseudo-State.

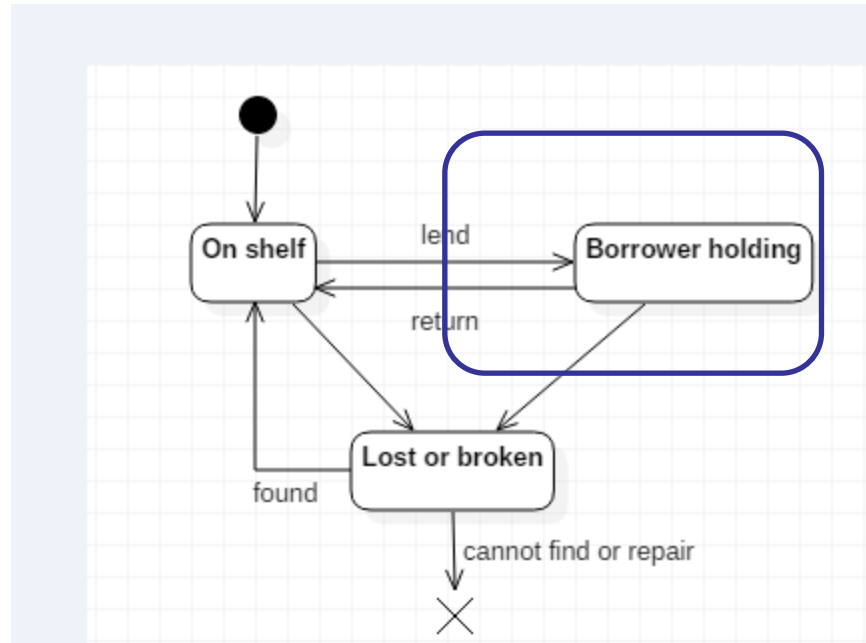
# What's inappropriate in the diagram??



# Suggestion

- In a state machine diagram, there is usually only one initial state.

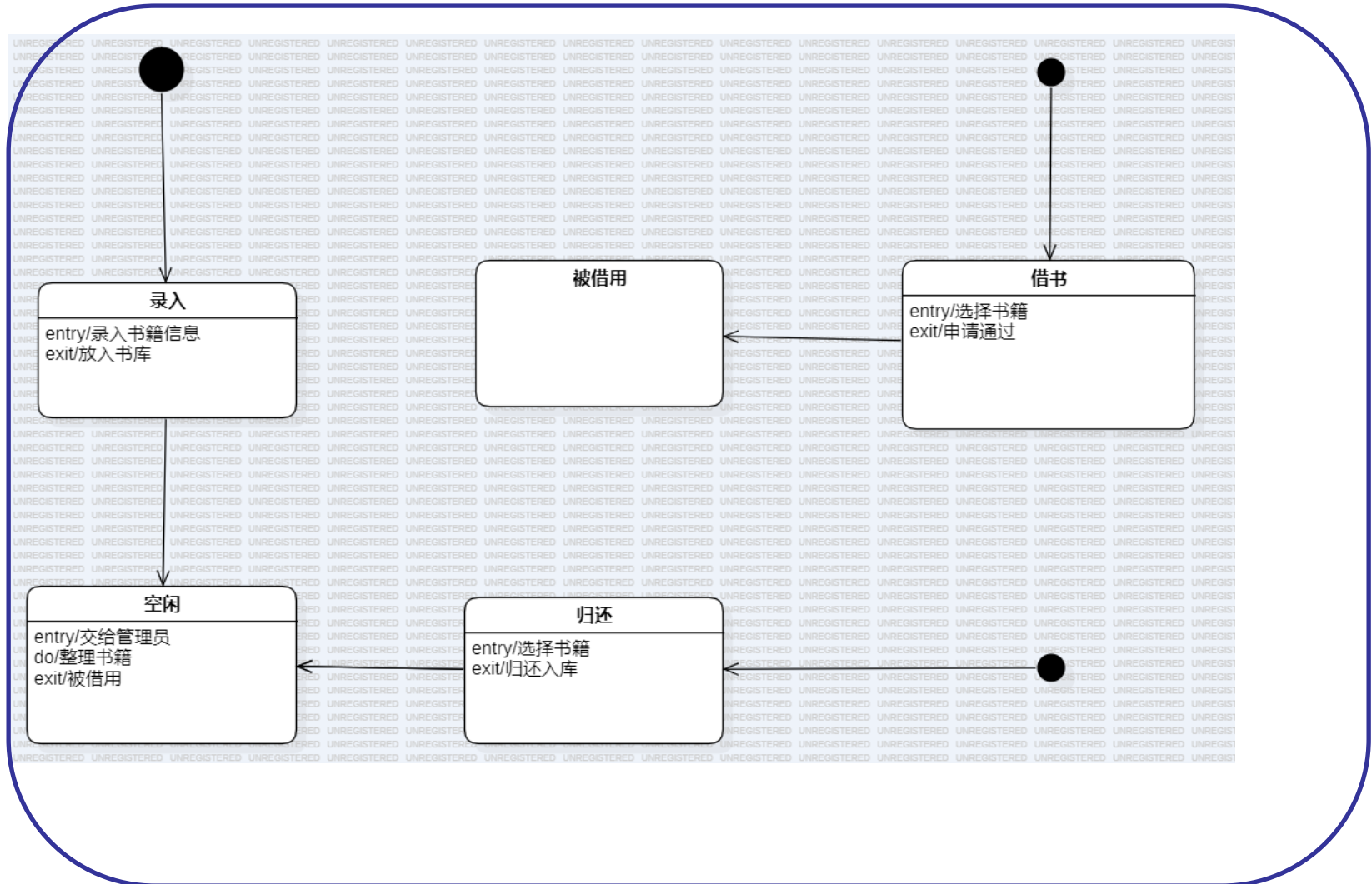
# What's inappropriate in the diagram??



# Suggestion

- When a state has multiple outgoing transitions, it is crucial to specify different triggering events for clarity.

# What's inappropriate in the diagram??

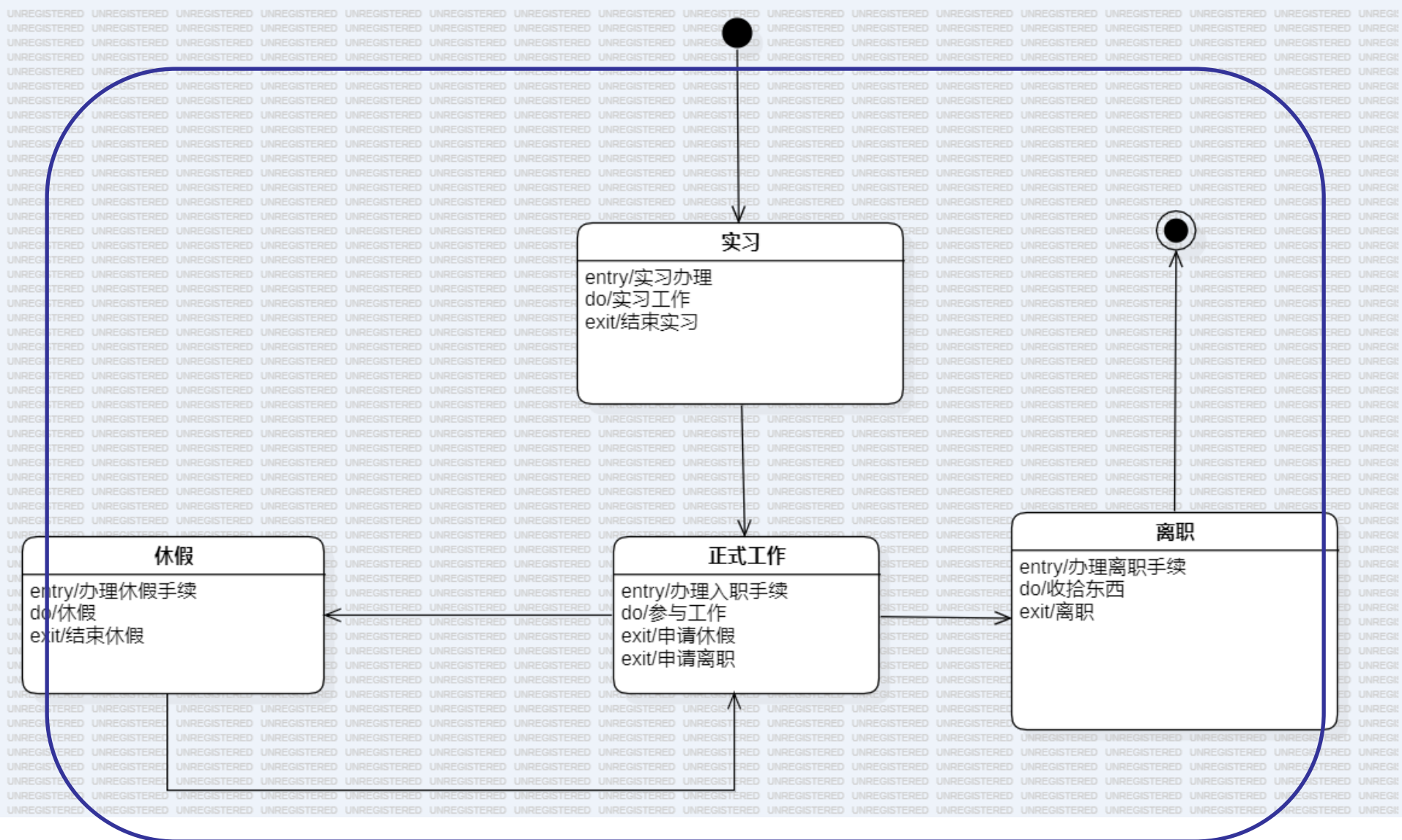


# Suggestion

- In a state machine diagram, nodes should remain connected.



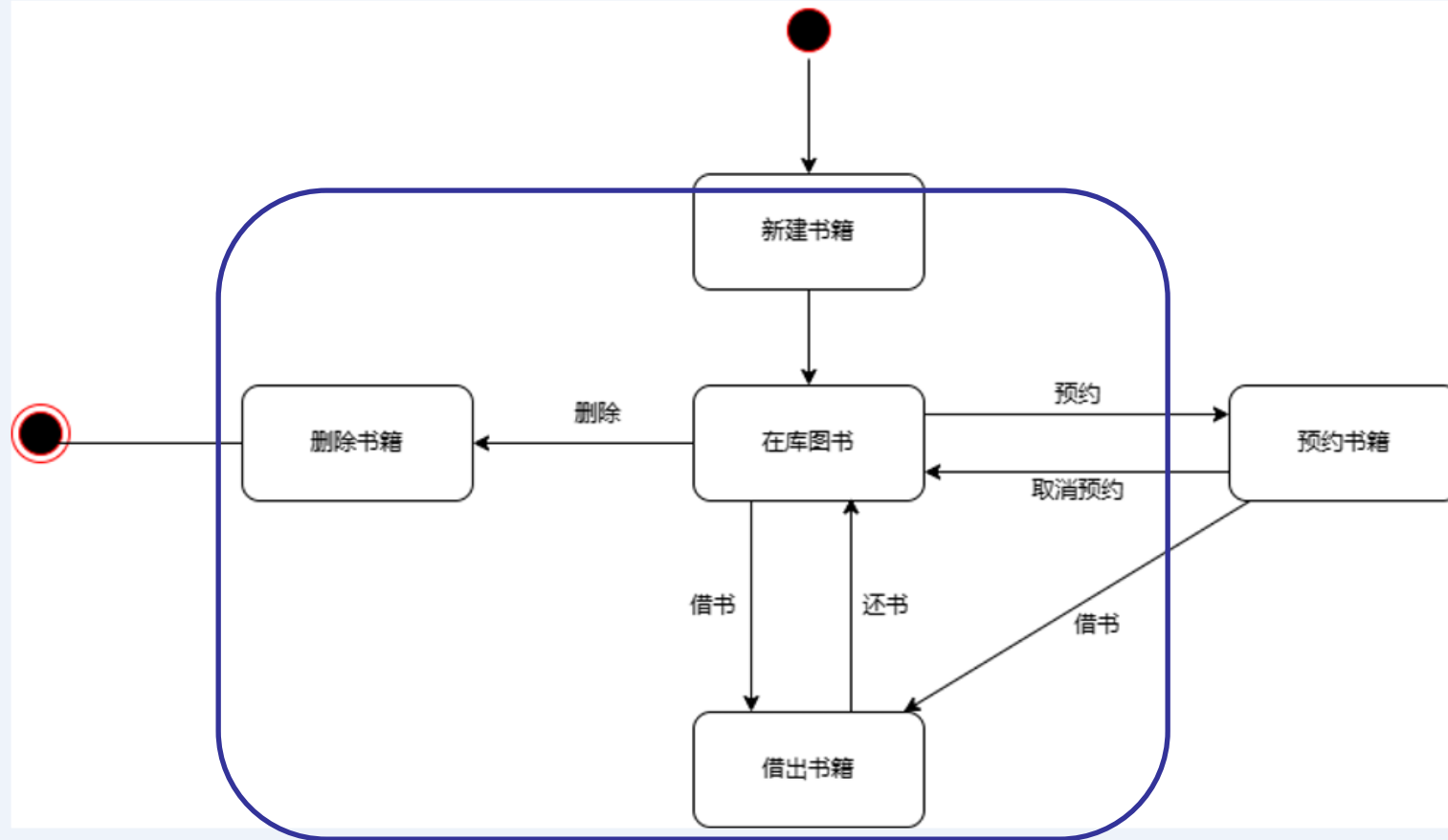
# What's inappropriate in the diagram??



# Suggestion

- State machine diagrams are distinct from activity diagrams; transitions between nodes should be event-triggered, and generally, unconditional transitions are not common.

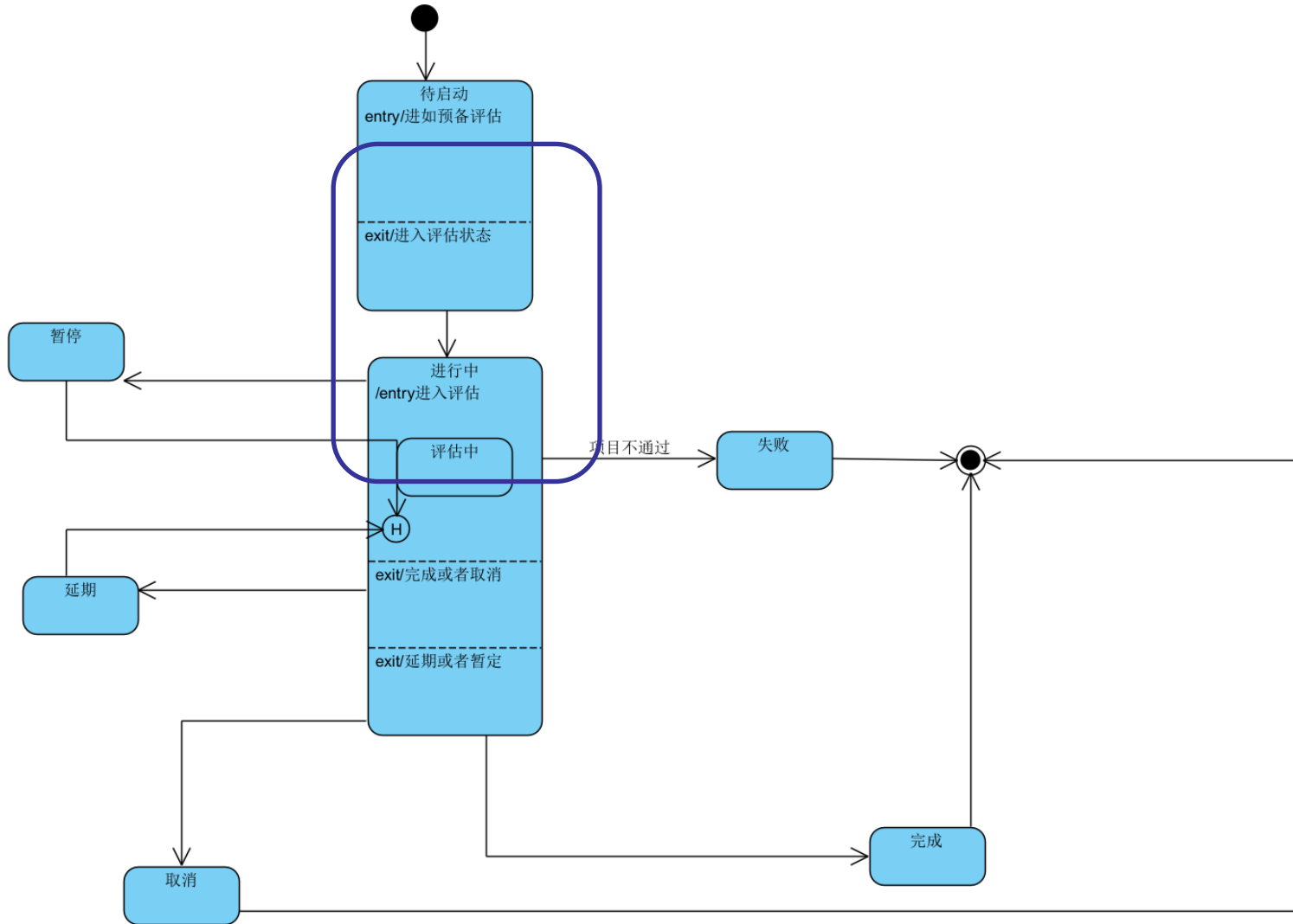
# What's inappropriate in the diagram??



# Suggestion

- Please pay attention to the naming of states in the state machine.

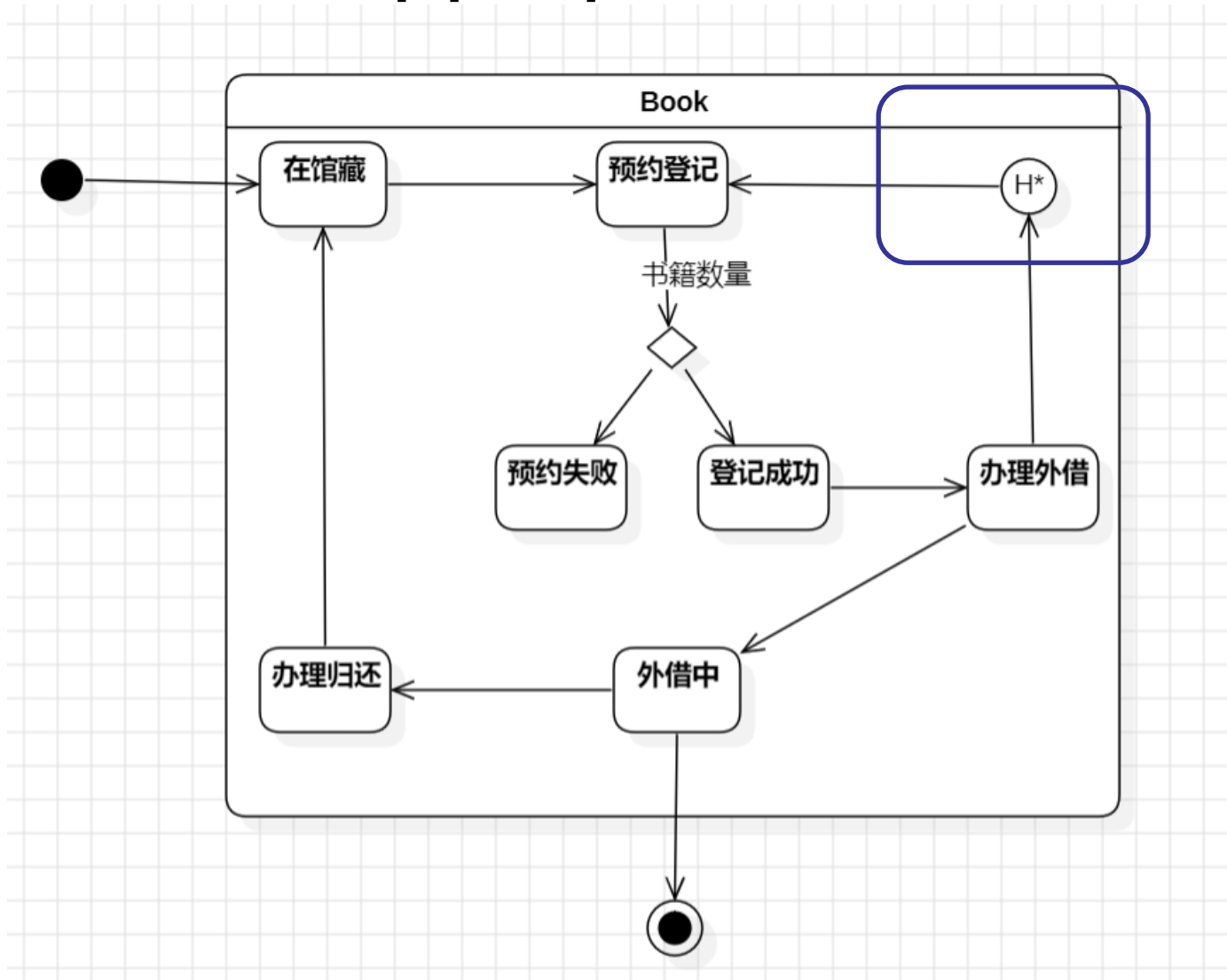
# What's inappropriate in the diagram??



# Suggestion

- Please pay attention to modeling the exit activities of states in a reasonable manner.

# What's inappropriate in the diagram??

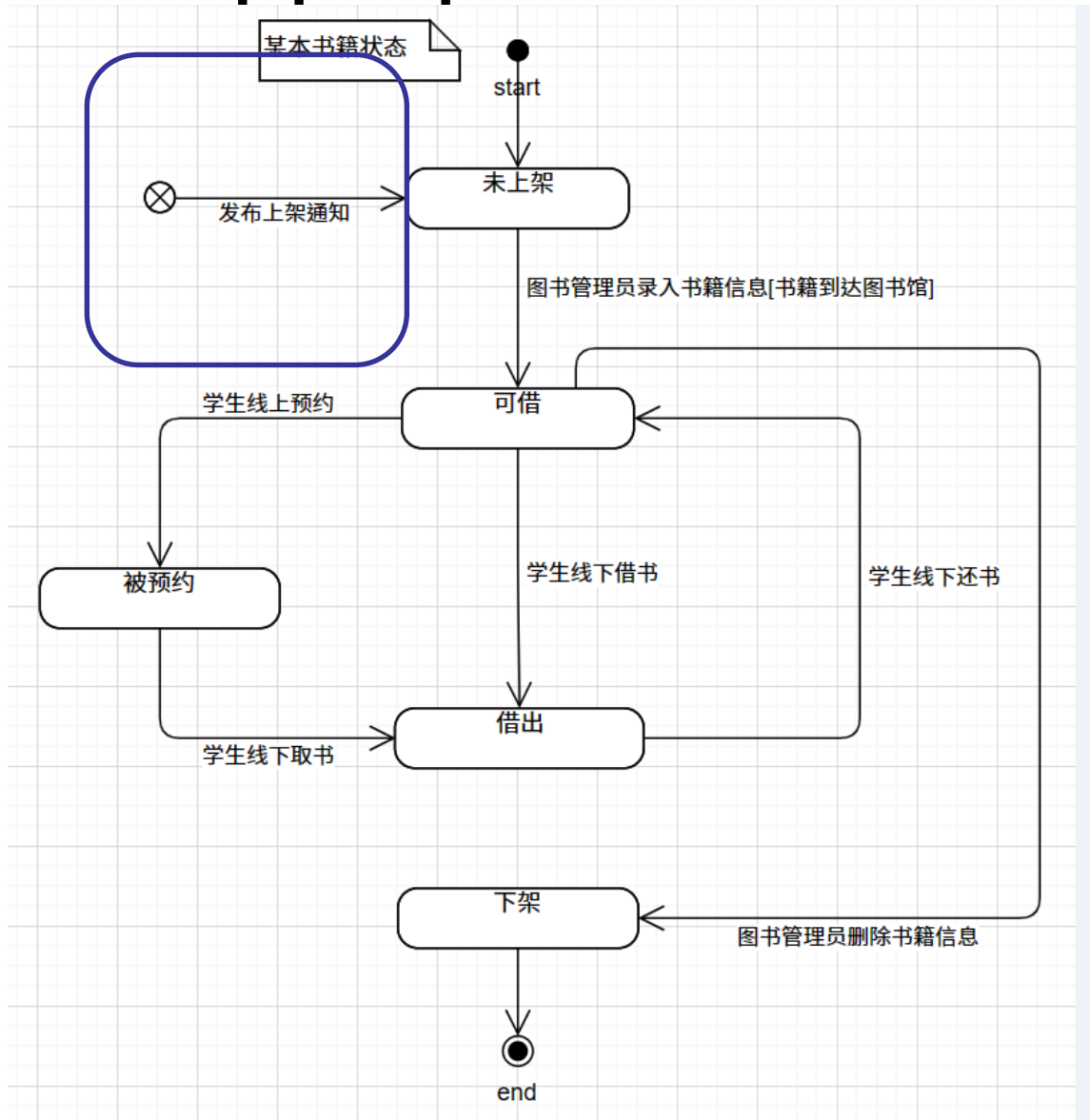


# Suggestion

- Please pay attention to the use of history states.



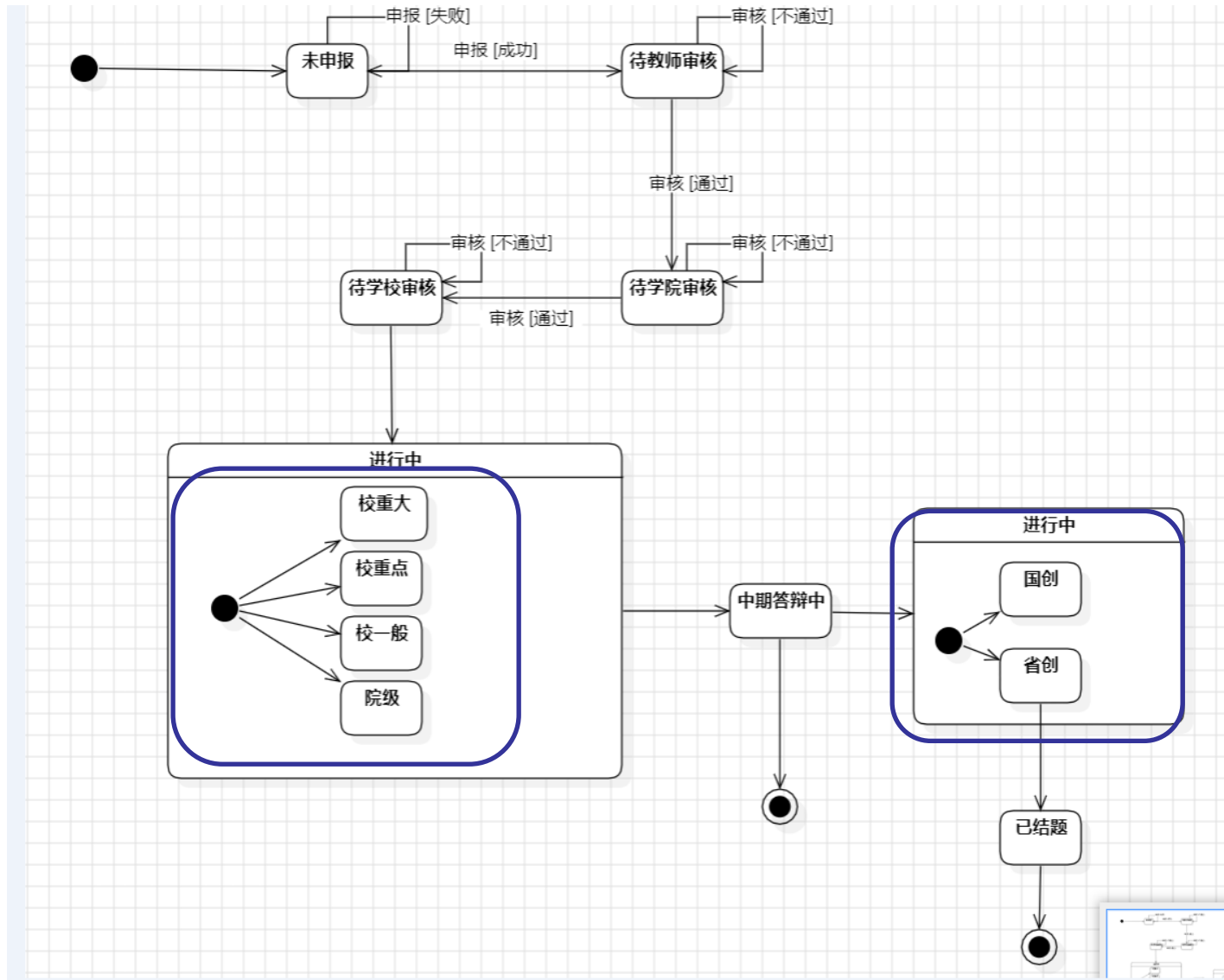
# What's inappropriate in the diagram??



# Suggestion

- Please pay attention to the reasonable use of the Termination State.

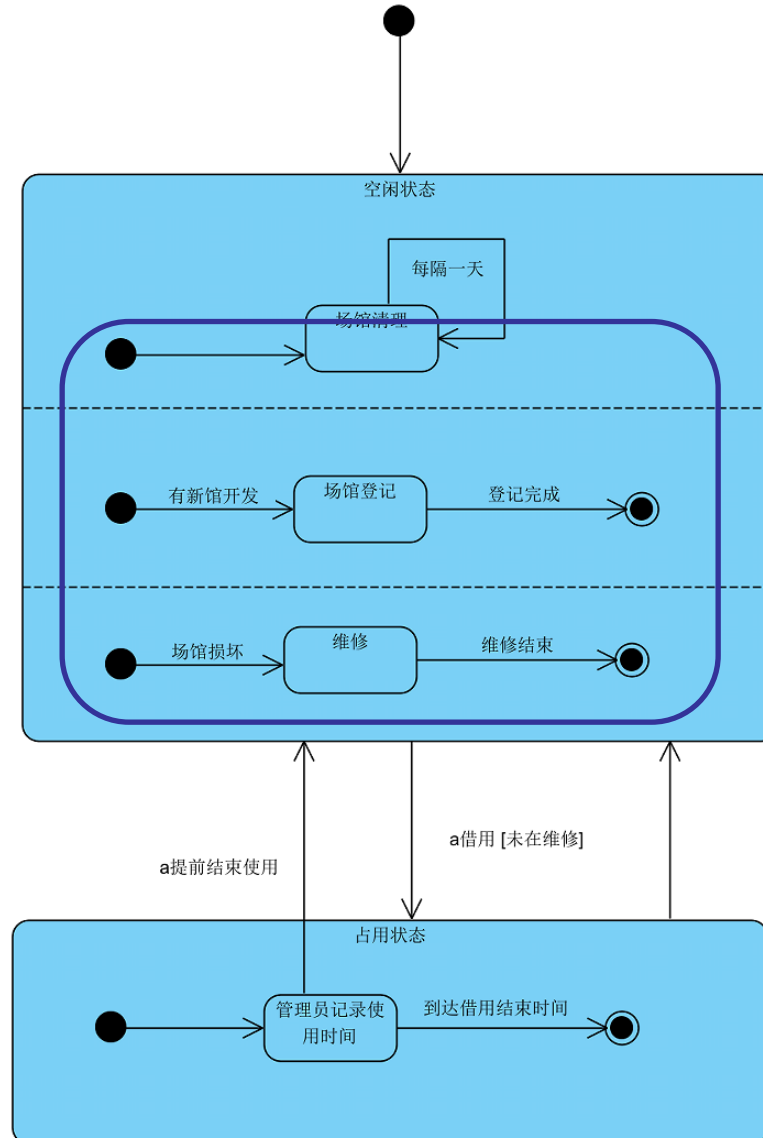
# What's inappropriate in the diagram??



# Suggestion

- Pay attention to transitions leaving the initial state.

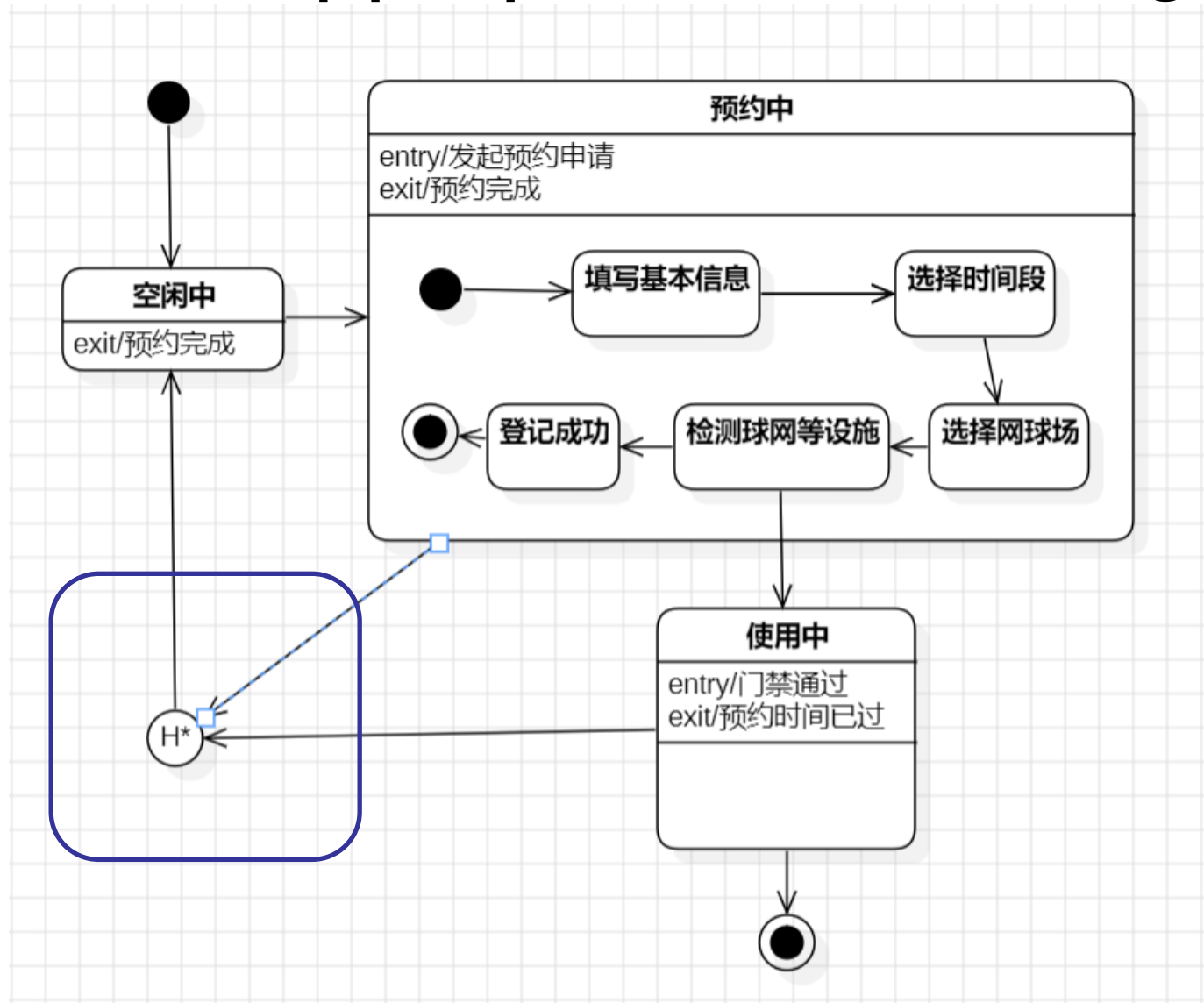
# What's inappropriate in the diagram??



# Suggestion

- Model substates within concurrent composite states appropriately.

# What's inappropriate in the diagram??



# Suggestion

- Please pay attention to the representation and usage of historical states.



# Suggestion

- Pay attention to the representation in the state machine diagrams
- Pay attention to the initial and final state
- Pay attention to the extraction of state
- Pay attention to the use of history state
- Pay attention to the composite state, and the composite state could have a name
- Pay attention to the transition between states
- Pay attention to the event on the transition
- Pay attention to the real meaning of the transition
- Pay attention to the expression of event
- Pay attention to the guard conditions
- Pay attention to difference between the event and guard condition